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Project Title	AI-Based Loan Approval Prediction System
Week	Week 2

WEEK-2 TASK

Task Name

AI-Based Loan Approval Prediction System

Abstract

An AI-Based Loan Approval Prediction System uses machine learning algorithms to predict whether a loan application should be approved or rejected. The system analyzes applicant details such as income, credit history, loan amount, employment status, and other financial factors. It helps banks and financial institutions make faster, accurate, and data-driven loan decisions while reducing manual effort and risk.

Introduction

Traditional loan approval processes require manual verification and may take more time. By using Artificial Intelligence and Machine Learning, the loan approval process can be automated. The system learns from previous loan data and predicts the probability of approval for new applicants.

Objectives

1. To automate the loan approval decision process.
2. To predict loan approval using machine learning models.
3. To reduce human errors and processing time.
4. To improve accuracy in credit risk assessment.
5. To assist banks in making better financial decisions.

Python code

```
from sklearn.model_selection  
  
import train_test_split
```

```
from sklearn.linear_model
import LogisticRegression
from sklearn.metrics
import accuracy_score

X = [[50000,200000,1],[30000,150000,0],[70000,250000,1]]
y = [1,0,1]

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.3, random_state=42)

model = LogisticRegression()
model.fit(X_train, y_train)
prediction = model.predict(X_test)

print("Prediction:", prediction)
print("Accuracy:", accuracy_score(y_test, prediction)).
```

Simulation Results & Observations

Prediction Efficiency:

The model successfully identifies patterns from applicant details and predicts loan approval decisions.

Accuracy:

Machine learning algorithms improve decision accuracy compared to traditional manual verification.

Benefits:

- Faster loan processing
- Reduced human errors

- Better risk assessment
- Automated decision-making

Conclusion

- The AI-Based Loan Approval Prediction System successfully demonstrates the use of machine learning for automated credit decision-making. The system helps financial institutions provide faster and more reliable loan approval services while reducing processing time and financial risks.